UNIVERSITI PENDIDIKAN SULTAN IDRIS

اونيۏرسيتي ڤنديديقن سلطان ادريس

SULTAN IDRIS EDUCATION UNIVERSITY

Software Verification and Validation (DES3103)

by

Dr. Nor Asiah Mohamad@Razak

Course Learning Outline

1. Briefing Session (Ice Breaking via Whiteboard, Class rules & Pro forma): **M. Teams**
2. Introduction to Software Verification & Validation I & II (V&V)
3. Managing V&V with SQA Integration
4. Quality metrics or attributes in V&V

WEEK 6-MID-TERM TEST (QUIZ): Topic 1 – 3-(CLO1:C2) – 20 MARKS

WEEK 7-PHASE 1: ASSIGNMENT 1-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO3:A5) - 20 MARKS: M. Teams

WEEK 9-PHASE 1: PRESENTATION & REFLECTION ON PHASE 1: M. Teams

5. Software Reviews – Will be covered on Assignments 1 & 2
6. Quality Control: Software Testing - Will be covered in Assignments 1 & 2
7. Human-Computer Interface Testing - Will be covered in Assignments 1 & 2
8. Manual Testing and Automation Testing: Will be covered in Final Assessment

WEEK 11-PHASE 2: ASSIGNMENT 2-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO2&4:P4&C5) - 20 MARKS: M. Teams
WEEK 13-PHASE 2: PRESENTATION & REFLECTION ON PHASE 2: M. Teams

9. Quality Maturity Models
10. Software Problem Analysis and Reporting
11. Quality Culture in doing V&V.
12. Safety and Security in V&V

WEEK 14: REFLECTION ON IMPLEMENTATION OPBL (SOFTWARE TESTERS) IN THE V&V COURSE: e-FORUM

WEEK 16-PHASE 3: FINAL ASSESSMENT – INDIVIDUAL SPECIAL PROJECT BASED ON PHASE 2 (CLO4:C5) (2 WEEKS) - 40 MARKS : e-ASSESSMENT

TOPIC 4

UNIVERSITI PENDIDIKAN SULTAN IDRIS
اونيۏرسيتي ڤنديديقن سلطان ادريس
SULTAN IDRIS EDUCATION UNIVERSITY

Quality metrics or attributes in V&V

- Quality metrics or attributes in V&V
- Importance of metrics
- Process versus artifacts metrics
- Defect metrics and variance
- Product metrics (quality attributes)
- Process metrics (process improvement) planning and choosing metrics, sample artifacts

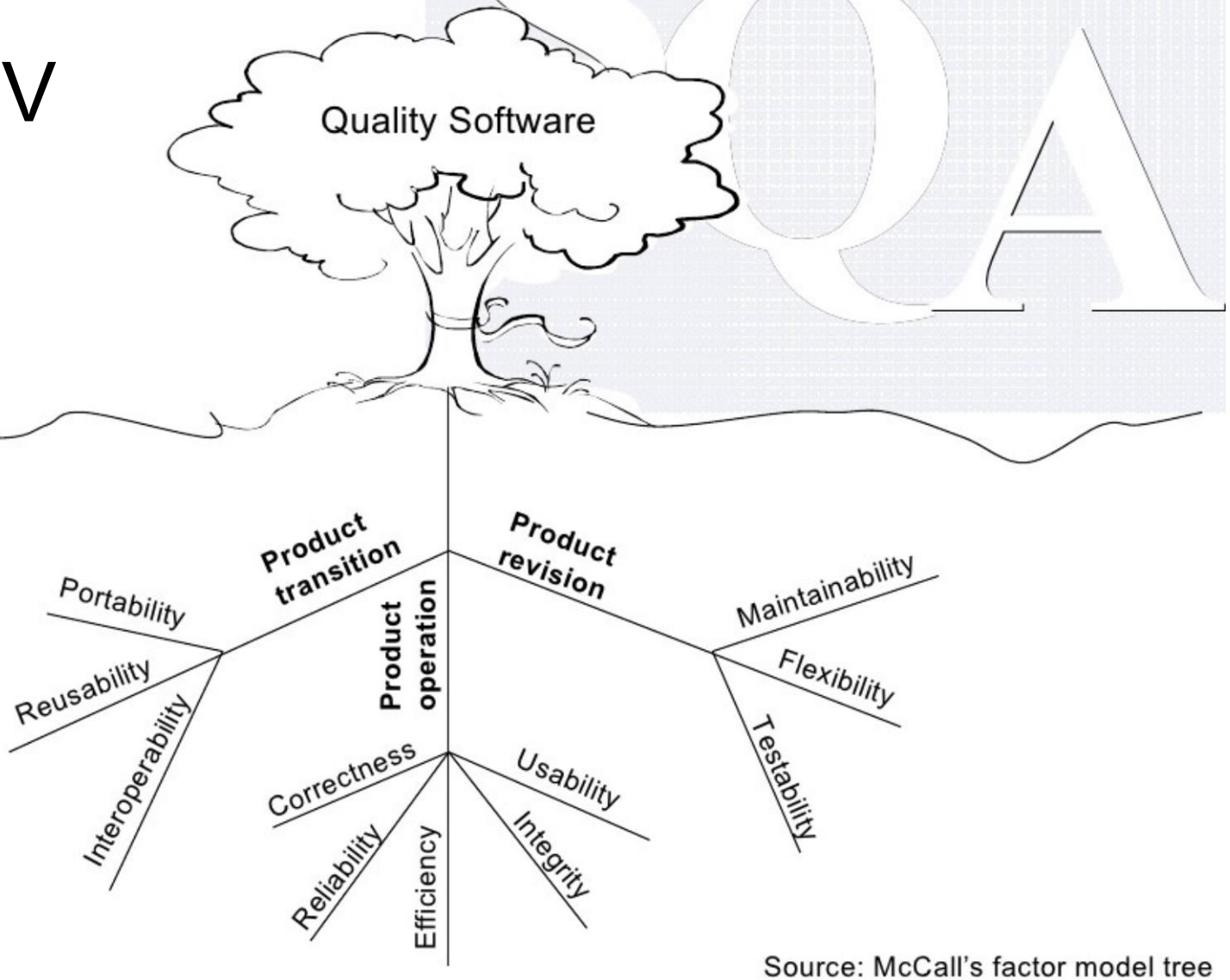
Roles of SQA in V&V

IEEE/ISO 9000 Definition of SQA

- Software Quality Assurance is: a systematic, planned set of actions necessary to provide adequate confidence that the software development process or the maintenance process of a software system product conforms to established functional technical requirements as well as with the managerial requirements of keeping the schedule and operation within the budgetary confines.

What is V & V and how does it fit with SQA?

- Verification ensures that "you built it right."
- Validation ensures that "you build what the customer asked for."
- V & V laid the foundation for SQA



Source: McCall's factor model tree

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

McCall's requirement factor category	Factor	Description
1. Product transition	Portability	Tend to the adaptation of a SW system to other environments consisting of: Different HW and Different OS Example: SW designed to work under Windows 10 is required to allow low-cost transfer to Linux.
	Reusability	Deals with the use of SW modules originally designed for one project in a new SW project currently begin developed. The reuse of SW is expected to save resources., shorten the project period, and provide higher quality modules. These benefits of higher quality are based on the assumption that most SW **faults have already been detected by SQA activities** performed previously on it.
	Interoperability	Focus on creating interfaces with other SW systems or with other equipment firmware. Example (save data storage): The firmware of medical lab. Equipment is required to process its results according to a standard data structure that can be then serve as input for a number of standard laboratory IS.

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

McCall's requirement factor category	Factor	Description
1. Product operation	Correctness	Correctness of the output. For example, the probability of non accurate inventory information, containing one or more mistakes, will not exceed 1%.
	Reliability	Deals with failures to provide service. E.g. One requirement of the new software system to be installed in the main branch of the Independence Bank, which operates 120 branches, is that on average, it will not fail more than 1 time per 12 months during the bank's office hours. In addition, the probability that the off-time (time needed for repair and recovery of all the bank's services) be more than 10 minutes is required to be less than 0.5%.
	Efficiency	Deals with the HW resources needed to perform. E.g. An outdoor meteorological unit. The system performs measurements once per hour, logs the results, and transmits the results once a day to the meteorological center by means of wireless communication. The unitise quipped with a 1,000 milliampere hour cell, which is capable of supplying the power requirements of the unit for at least 30 days.
	Integrity	Deals with the SW system security, that is requirements to prevent access to unauthorized persons.
	Usability	Deals with the scope of staff resources needed to train a new employee and to operate the SW system. (1) A staff member should be able to handle at least 60 service calls a day.

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

McCall's requirement factor category	Factor	Description
1. Product revision	Maintainability	Requirements determine the efforts that will be needed by users and maintenance personnel to identify the reasons for SW failures, to correct the failure, and to verify the success of the corrections. Example: Typical maintainability requirements: 1. The size of a SW module will not exceed 30 statements 2. The programming will adhere to the company coding standards and guidelines.
	Flexibility	The capabilities and efforts required to support adaptive maintenance activities are covered by flexibility requirements. This factor's requirements also support perfective maintenance activities, such as changes and additions to the SW to improve its service and adapt it to changes in the firm's technical or commercial environment.
	Testability	Deal with the testing of an IS as well as with its operation. Providing predefined intermediate results and log files. Automatic diagnostics performed by the SW system prior starting the system, to find out whether all components of SW system are in working order. Obtain a report about detected faults.

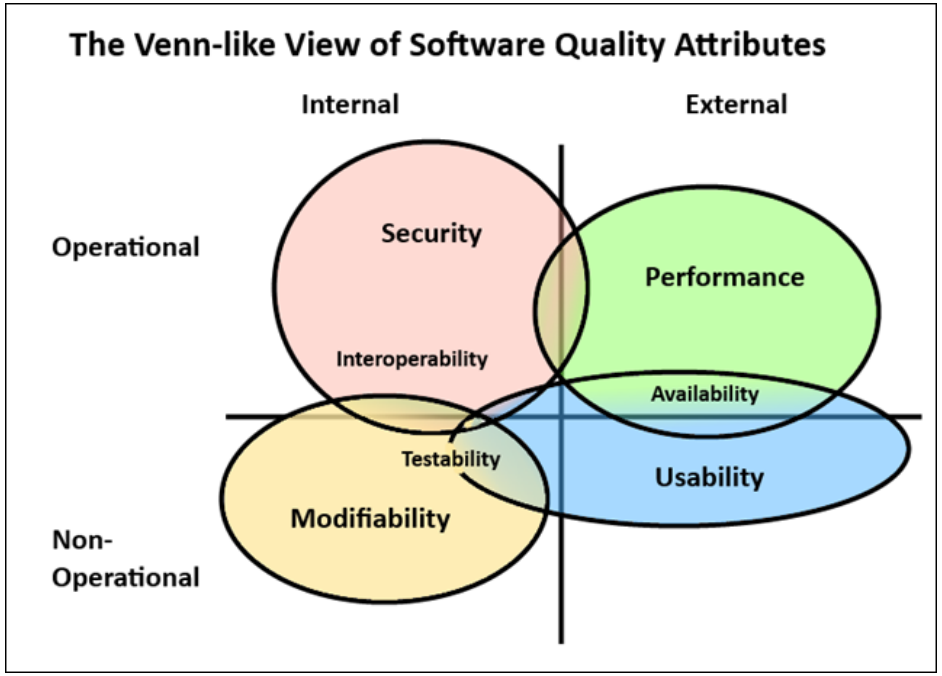
Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

What are the software quality attributes?

- Quality attribute requirements are part of an application's nonfunctional requirements, which capture the many facets of how the functional requirements of an application are achieved. All but the most trivial application will have nonfunctional requirements that can be expressed in terms of quality attribute requirements.
- The nonfunctional requirements define these aspects about the system. (The nonfunctional requirements are sometimes referred to as "nonbehavioral requirements" or "software quality attributes.")
- Non-functional requirements, as the name suggests, are requirements that are not directly concerned with the specific services delivered by the system to its users. They may relate to emergent system properties such as reliability, response time, and store occupancy. Alternatively, they may define constraints on the system implementation such as the capabilities of I/O devices or the data representations used in interfaces with other systems.

Most quality attributes will fit into one or two of the big; four: *security*, *performance*, *usability* and *modifiability*. And each of these largely corresponds to its depicted quadrant.

Security is largely an internal concern. Often external stakeholders will profess their concern with this attribute, and hence the Venn-like spill over into the external quadrant. Generally the concern for security and assuring security falls within the domain of internal stakeholders and the implementation team.



External stakeholders are mostly concerned with performance and usability. There are a number of other quality attributes such as availability that fall within or at least are closely related to one or both of those.

Modifiability is almost exclusively a concern of internal stakeholders and implementation teams. Interoperability and testability are largely related to modifiability while interoperability nearly always shares at least some security concerns.

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

Requirements	Factor	McCall'sv requirement factor category
The probability a software system will be found in a state of failure during peak hours (9am-4pm) is required to be below 0.5%.	Reliability	Product operation
The software package developed for the Linux operating system should be compatible for applications in a Windows NT environment.	Portability	Product transition

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

Functional and non-functional requirements

When it comes to quality in software development, there are functional and non-functional requirements:

- **Functional requirements** specify what a system/product should do. (Example: A store system should provide the customer with an overview of what they have already purchased.)
- **Non-functional requirements** go beyond functional requirements and specify how well a system/product should perform a function or service. (Example: The system should display the purchase overview within three seconds.) Non-functional requirements are also known as boundary conditions or quality attributes.

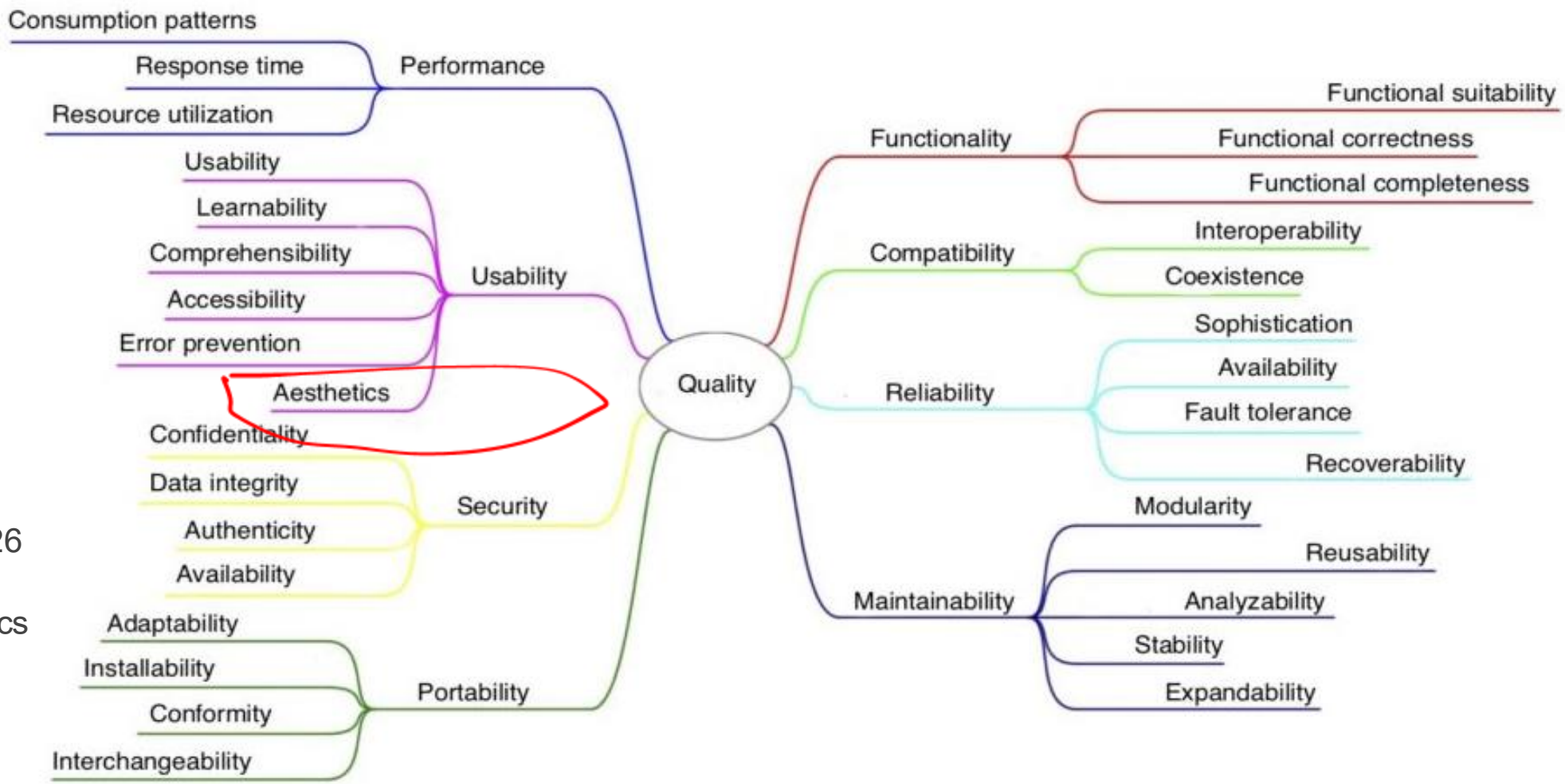
Source / Feature	Safety Regulation/ Regulatory Guide	IEEE Industry Standarts	Configuration Management Process Area
Function	- Specific requirement on protection system - Endorse IEEE industry standarts - Reflect to standarts review plan	- Specific SCM activities - Change management - Supplier control - Tools, techniques and methodologies - Provide some samples plan	- Specific and generic goal - Specific and generic practice - Refer to other process area
Non-function	- Quality - Reliability - Quality assurance - Manage configuration activities	- Traceability - Consistency - No specific descriptions for safety system	- Integrity - Support other process area - No specific descriptions for safety system

Source: Research Gate

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

Classification of quality characteristics

Here are **other samples of quality attributes**, what are the most common requirements for them to achieve them.



The ISO-25010/9126 standard classifies quality characteristics as follows:

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

What are the software quality attributes?

Other Quality Attributes

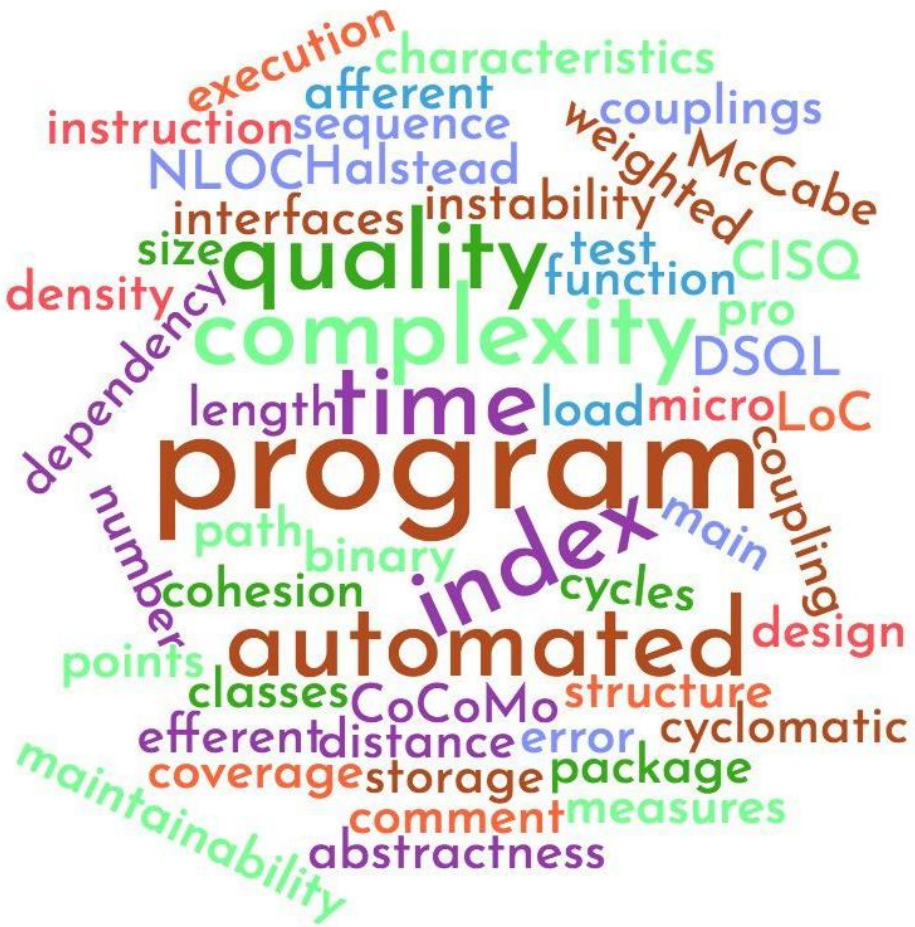
The standard quality attribute list for product quality

The ISO/IEC FCD 25010 product quality standard

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

Metrics

There are various metrics used to assess these characteristics. This word cloud presents a selection: And that is by no means all of them.



Methods and tools

Software teams have high-performance tools and concepts at their disposal for all of the aspects mentioned. Here are a few examples:

- Maintainability/Portability:
SonarQube, evaluation processes - Architecture Tradeoff Analysis Method (ATAM, audits)

- Security:
OWASP CVE Scanner, Jess Security Scanner, SonarQube : https://www.sonarqube.org

- Reliability:
Stress tests, regression tests

- Usability:
A/B tests, WAI/AAA tests, usability inspections (guideline-based reviews), surveys

- Performance:
Load tests, performance tests

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

Common Quality Attributes examples

Here are some samples of quality attributes, what are the most common requirements for them, also some metrics and strategies to achieve them.

Availability

Availability defines the proportion of time that the system is functional and working.

It can be measured as a percentage of the total system downtime over a predefined period.

System errors, infrastructure problems, malicious attacks, and system load will affect overall system availability.

Requirements	The system should be highly available
General scenario	The system should have 99.9% available time per month with maximum monthly 43m 49.7s downtime The system should notify users if there is an issue affects the availability or may cause downtime. In case of upgrade or maintenance, the maintenance window will be out of business hours.
Metrics	The equation below is used to calculate the availability $A = \frac{MTBF}{MTBF + MTTR}$ A: Availability MTBF: Mean Time Between Failures MTTR: Mean Time to RepairOur target of this metrics to ensure that we will not have more 43 minutes downtime per month. Our target of this metrics to ensure that we will not have more 43 minutes of downtime per month.
Strategies to Achieve the quality metrics	Using cloud technologies based on different region as data center redundancy pattern. Using redundant infrastructure components. Avail availability status page, which our users can subscribe to receive availability status reports and incidents.
Side effects	The cost of services will be higher to have a high available environment and have redundant hardware and licenses. It also has trade-offs with other quality attributes, for example, reliability.

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

Common Quality Attributes examples

Here are some samples of quality attributes, what are the most common requirements for them, also some metrics and strategies to achieve them.

Usability

Usability is the degree to which a software can be used by specified consumers to achieve quantified objectives with effectiveness, efficiency, and satisfaction in a quantified context of use, for example, easy to localize and globalize, providing good access for disabled users, and resulting in a good overall user experience.

Requirements	The system should be user-friendly A new user can reach proficiency for core functions in a week.
General scenario	The solution will be designed to have enrollment steps for new users to introduce the system to them. The number of mistakes by new users will be counted to measure was it easy for these users to do a particular process or function easily or not. The system will have a help manual to support users to know how to do a specific function.
Metrics	Completion rate: the rate of users who completed a task successfully Number of errors: the number of errors the user faced while doing a task on the system The duration time: the duration, the user spent to finish the task Training: the number of days required to train new users User satisfaction surveys
Strategies to Achieve the quality metrics	The solution will use auto-completion algorithm to help the user fill the data easily. Any forms will take more than 5 minutes will be split into more than one form. The navigation buttons will be designed according to usability standards. Using behavioral analytics techniques to analyze user behavior. A/B testing.
Side effects	Personalization and Usability are interdependent, some personalization requirements can be generated, for example, context-aware applications which will affect usability as well. The behavior data will be generated and collected at the runtime, which needs extra processing, storage, extra effort for visualizing and analyzing this data.

Quality metrics or attributes in V&V - Quality metrics or attributes in V&V

Common Quality Attributes examples

Here are some samples of quality attributes, what are the most common requirements for them, also some metrics and strategies to achieve them.

Performance

The degree to which a system or component accomplishes its designated functions within given constraints, such as speed, accuracy, or memory usage. It refers to responsiveness, either the time required to respond to specific events, or the number of events processed in a given interval of time.

Requirements	The system should respond to a request or event in less than 10 seconds. The system should able to handle 200 users.
General scenario	The solution will be designed to handle 200 users concurrently with the average response time of 10 seconds. Caching techniques will be used to enhance response time for the same type of queries.
Metrics	Latency: is the time taken to respond to any event Throughput: is the number of events that take place within a given amount of time Capacity: is the total demands can be handled by the system while continuing to meet latency and throughput requirements Modes: having different resources or demand over time and system adaptation to that.
Strategies to Achieve the quality metrics	There are different strategies to enhance system performance, for example, caching techniques, data partitioning, load balancing, scalability, First Things First (FTF)
Side effects	Performance has a lot of tradeoffs, and relation with other quality attributes, for example, dependability. Also, special testing procedures shall be available to test the performance, for example, load testing.

It would be helpful as well to look at McCall's and Boehm's Quality Models. There are tens of quality attributes, and you can find a good list in this ***link****.*

Quality metrics or attributes in V&V - process metrics (process improvement)

Verification Validation Maturity Model (V2M2)

The most vital factors for starting maturity assessment with V2M2 model are combined into a matrix (the Table) that provides a softer landing for the use of the actual model itself.

The applicability of the matrix is tested in the empirical part, where it is used as an aid for analyses.

Level 1 Initial	Level 2 Repeatable	Level 3 Defined	Level 4 Managed and Aligned	Level 5 Optimising
Main objective of V&V: *To show in an ad-hoc way that products work*	**Main objective of V&V:** *Defect detection*	**Main objective of V&V:** *Are based on the requirements*	**Main objective of V&V:** *To provide quantitative measures to allow visibility for management*	**Main objective of V&V:** *To act as a total product quality control, to result in low risk products*
Purpose of the level: N/A	**Purpose of the level:** *The establishment of a basic verification and validation process* *By the introduction of basic practices, basic V&V process emerges*	**Purpose of the level:** *To further organise V&V and to embed it into the development life-cycle* *A defined and repeatable process is in place, documented standards and procedures*	**Purpose of the level:** *The establishment of V&V as a real quality measurement* *By aligning the way-of-working with other organisational entities*	**Purpose of the level:** *To fine-tune and optimise the V&V process on a continuous and structured basis*
V&V is finding and correcting problems Lack of tools, resources and properly trained staff	V&V is performed in a systematic and planned way Work products are documented Tests are conducted in a dedicated V&V environment	Activities start already at the requirements phase and continue through the entire life-cycle A V&V organisation is in place and testing is recognised as a profession	Quantitative measurements, statistical techniques and methods control the process	Costs, efficiency and effectiveness are quantitatively measured Defect causal analysis and defect prevention is a common practice The V&V process is continuously monitored and improved
Process areas: None	**Process areas:** 1. V&V policy and goals 2. V&V project planning 3. V&V monitor and control 4. V&V design methodology 5. V&V environment	**Process areas:** 1. Organisation embedding 2. Training program 3. V&V life-cycle embedding 4. Peer reviews	**Process areas:** 1. Organisational alignment 2. Quality measurement and evaluation 3. Quantitative process management	**Process areas:** 1. Defect prevention 2. Quality management 3. Process optimisation

Maturity level matrix – based on Metric based Verification Validation Maturity Model (V2M2)

Quality metrics or attributes in V&V - process metrics (process improvement)

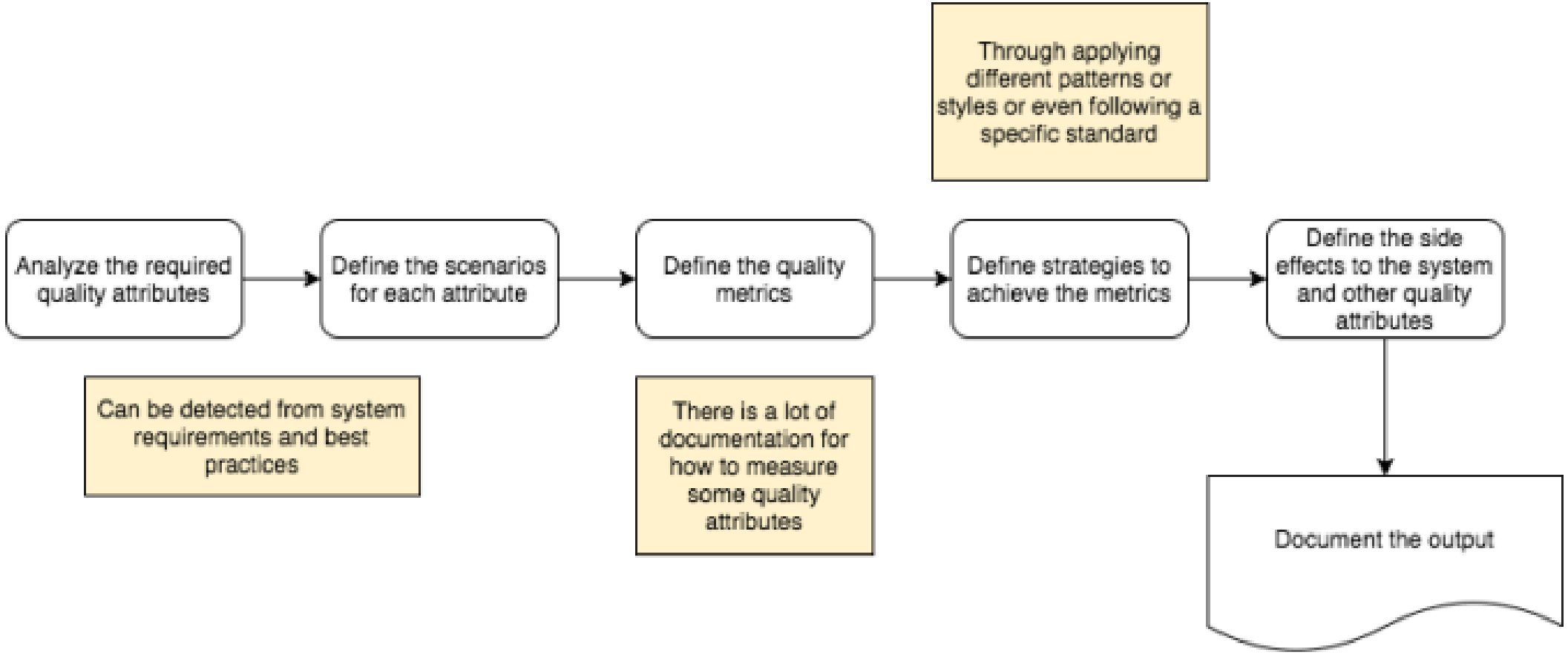
Process Steps for Planning V&V.

- Planning the V & V effort is a reoccurring task used to assure V & V resources are efficiently identified and allocated.
- Planning also establishes the goals and objectives for the V &V effort.
- We must emphasize that it is a reoccurring task and not merely performed at the beginning of the project to never be addressed again.
- Management, scientists, and engineers must work together through the course of the project to assure that realistic plans are established and maintained.
- It is not possible to plan a completely risk-free project.
- There are too many unknowns and too many variables to consider. As such, plans need to be reassessed during the life of the project.
- Planning can be organized int o a series of steps.
- These steps are identified in the Table.

Planning Process Step	Description
Establish V&V mission Identify V&V stakeholders	Develop the goal for the V &V project. Identify those entities that have a vested interest in the V&V project.
Identify V & V stakeholder requirements	Identify the needs and expectations of the stakeholders.
Establish the V & V objectives	Identify five to seven results that define a successful V&V project.
Develop a concept	Develop a high-level approach depicting how the V&V project will operate.
Develop V & V requirements	Using the stakeholder requirements, V & V concept, and the standard V & V requirements develop the system requirements for the V &V project.
Establish V & V scope	Using the V & V objectives, identify the software components that will be assessed during the project.

Quality metrics or attributes in V&V - process metrics (process improvement)

As we have a software development process, we should apply the software quality process to ensure the applicability and existence of the selected qualities. The drawing below process to analyze and plan for quality attributes, I think this can be integrated with software quality process and apply some checklist with each deliverable if the deliverable is fulfilling these qualities.



Please read for preparing your test

- https://www.softwaretestinghelp.com/how-to-write-test-plan-document-software-testing-training-day3/

END

UNIVERSITI PENDIDIKAN SULTAN IDRIS
اونيۏرسيتي ڤنديديقن سلطان ادريس
SULTAN IDRIS EDUCATION UNIVERSITY

Thank You…
Terima kasih..